

well as:

$$\text{mfence}, s \xrightarrow{\text{MF}}_c \text{skip}, s$$

TSO Storage transitions: $M, B \xrightarrow{\tau:l}_{\text{mf}} M', B'$.

$$\frac{B(\tau) = b \quad b' = b.(W, x, v) \quad B' = B[\tau \mapsto b']}{M, B \xrightarrow{\tau:(W, x, v)}_{\text{mf}} M, B'}$$

$$\frac{B(\tau) = b \quad \text{get}(M, b, x) = v}{M, B \xrightarrow{\tau:(R, x, v)}_{\text{mf}} M, B}$$

$$\text{get}(M, b, x) \triangleq \begin{cases} v & \text{if } \exists b_1, b_2. b = b_1.(W, x, v).b_2 \\ & \wedge \neg \exists v'. (W, x, v') \in b_2 \\ M(x) & \text{otherwise} \end{cases}$$

$$\frac{B(\tau) = \emptyset}{M, B \xrightarrow{\tau:\text{MF}}_c M, B}$$

$$\frac{B(\tau) = \emptyset \quad M(x) = v_o}{M, B \xrightarrow{\tau:(U, x, v_o, \perp)}_{\text{mf}} M, B}$$

$$\frac{B(\tau) = \emptyset \quad M(x) = v_o \quad M' = M[x \mapsto v_n]}{M, B \xrightarrow{\tau:(U, x, v_o, v_n)}_{\text{mf}} M', B}$$

$$\frac{B(\tau) = (W, x, v).b \quad M' = M[x \mapsto v] \quad B' = B[\tau \mapsto b]}{M, B \xrightarrow{\tau:\epsilon}_{\text{mf}} M', B'}$$